

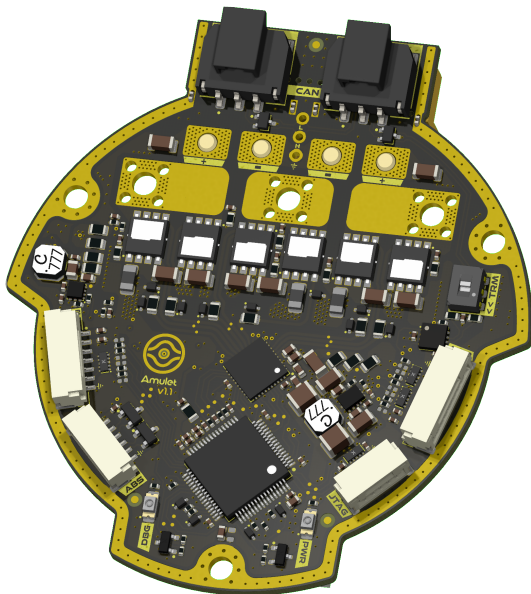
Amulet Motion Controller

Variant:

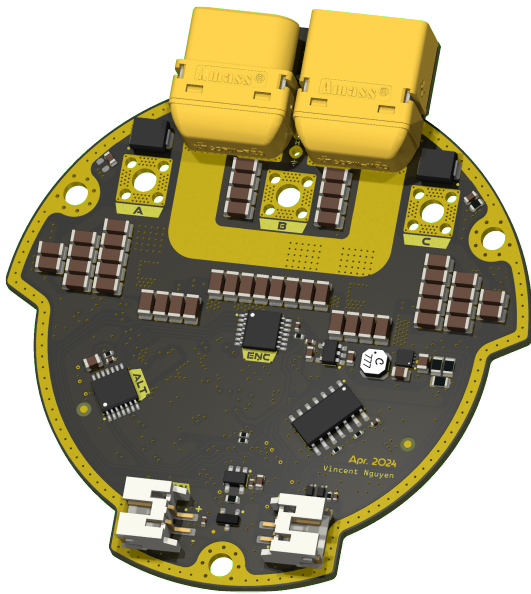
2024-11-21
Rev 1.1

Page	Index	Page	Index	Page	Index	Page	Index
1	Cover Page	11	User - LED Indicators	21	Revision History	31	
2	Block Diagram	12	Sensing - Temperature	22		32	
3	Project Architecture	13	Sensing - Battery	23		33	
4	MCU - Power	14	Sensing - Position	24		34	
5	MCU - I/Os	15	Interface - RS-422	25		35	
6	Power - Generation	16	Interface - FD-CAN	26		36	
7	Power - Connectors	17	Interface - Fan Control	27		37	
8	Motor Control - Top Level	18	Interface - Interconnects	28		38	
9	Motor Control - Inverter	19	Misc - Holes, Fiducials	29		39	
10	Misc - Board Version, DAC	20	Power - Sequencing	30		40	

TOP VIEW



BOTTOM VIEW



DESIGN CONSIDERATIONS

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

DESIGN NOTE:

Example text for critical design notes.

LAYOUT NOTE:

Example text for critical layout guidelines.

NOTES

Schematic based off Josh Pieper's moteus controllers.

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

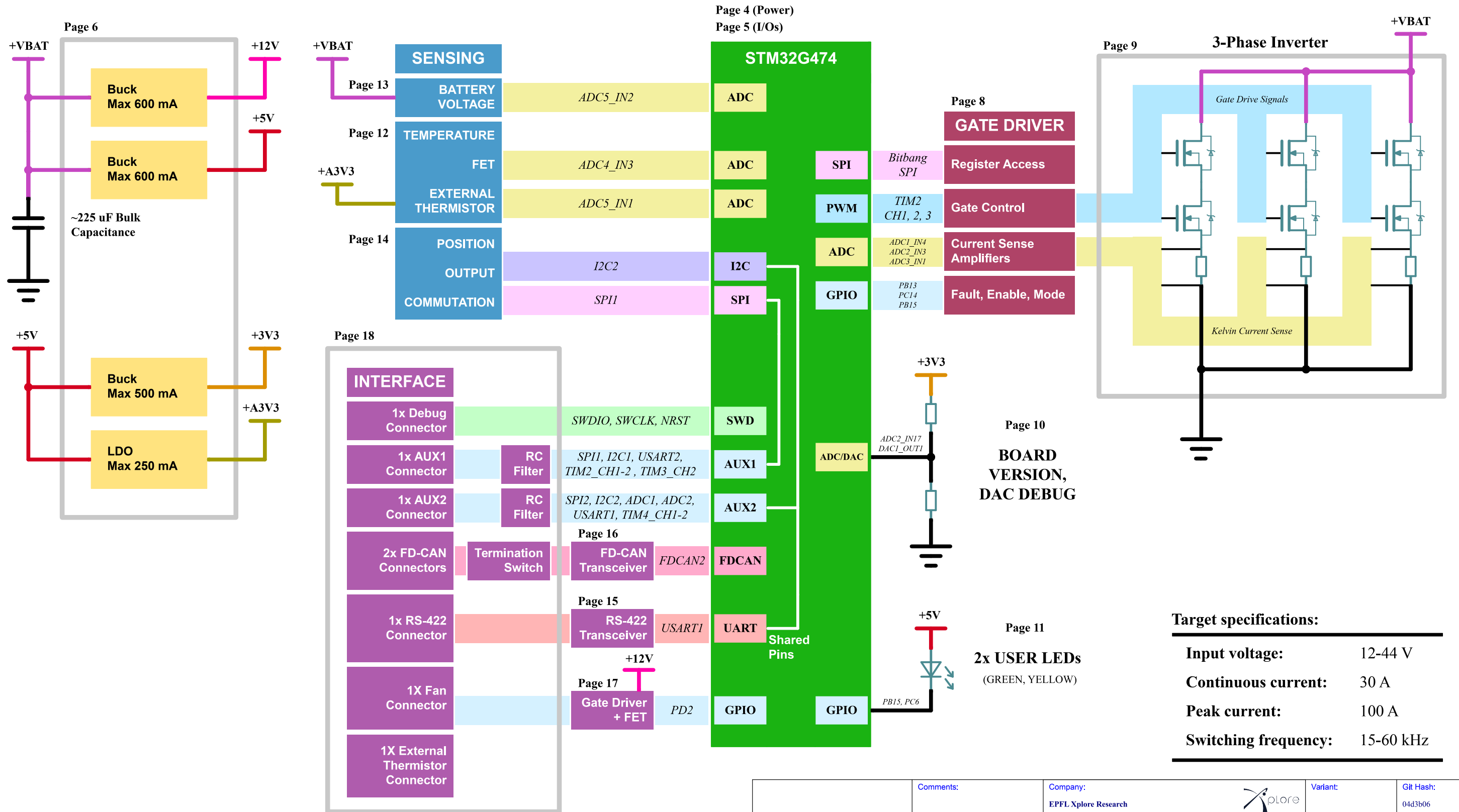
CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Date: 13-APR-2024

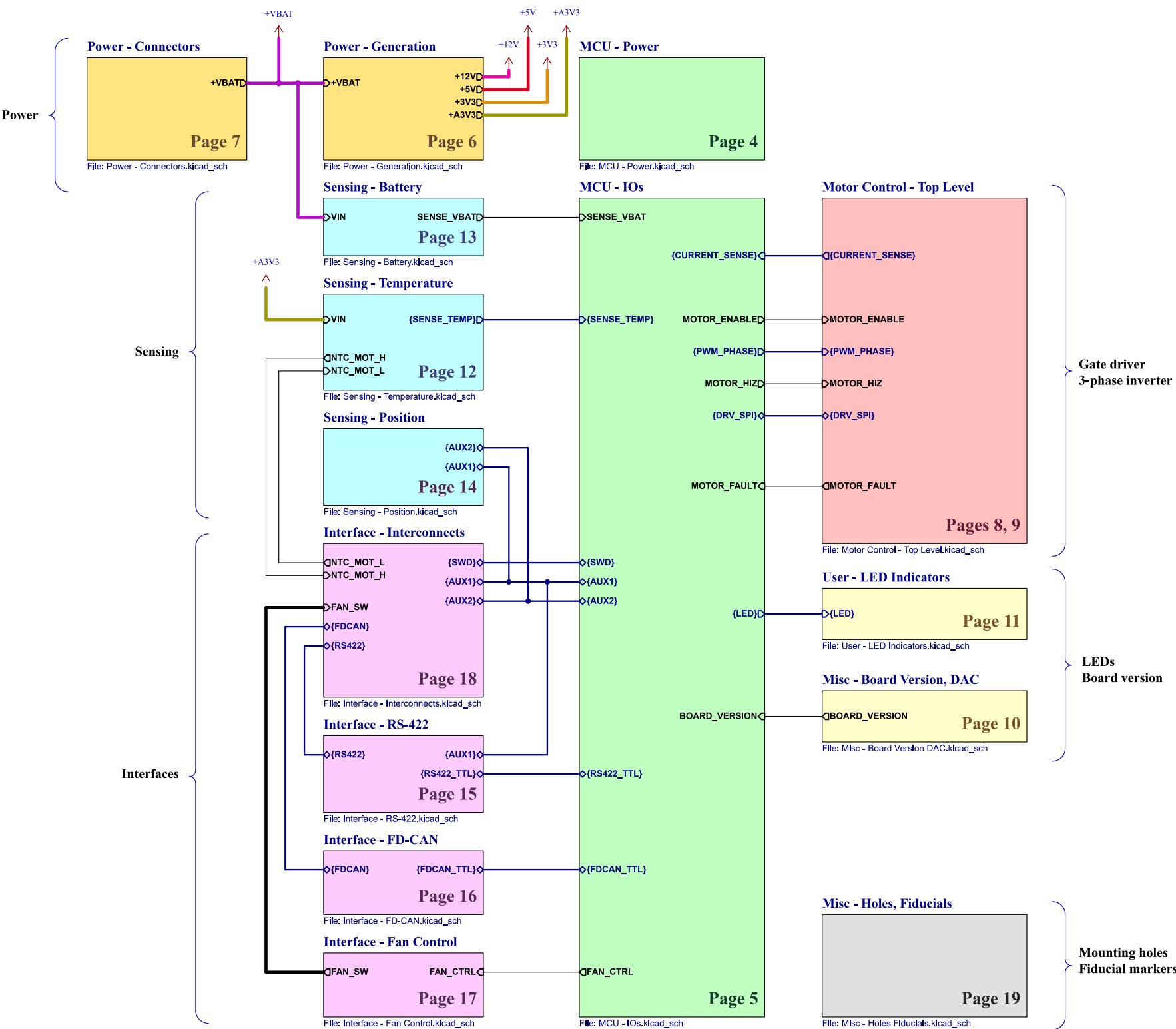
	Comments:	Company: EPFL Xplore Research			Variant:	Git Hash: 04d3b06
		Board Name: Amulet Motion Controller			Project Name: Chienpanzé	
	Sheet Title: Cover Page	File Name: amulet_controller.kicad_sch	Designer: Vincent Nguyen	Date: 2024-04-13	Revision: 1.1	
	Sheet Path: /		Reviewer:		Size: A3	Sheet: 1 of 21

[2] Block Diagram



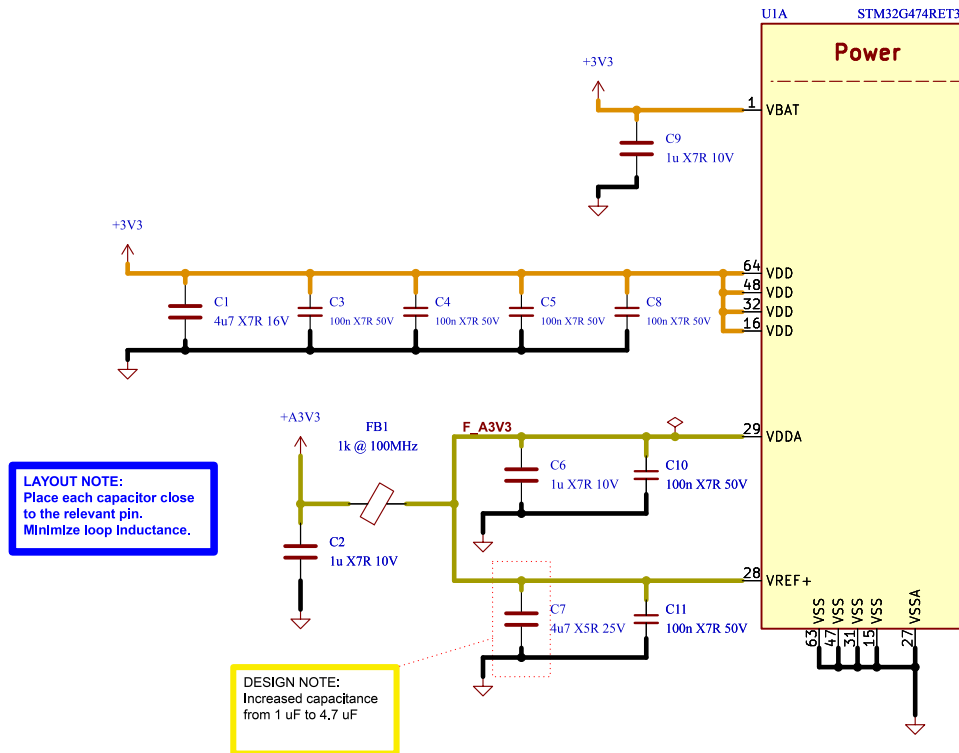
Comments:	Company: EPFL Xplore Research				Variant:	Git Hash: 04d3b06
	Board Name: Amulet Motion Controller			Project Name: Chienpanzé		
Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: Vincent Nguyen	Date: 2024-04-13	Revision: 1.1		
Sheet Path: /Block Diagram/		Reviewer:	Size: A3	Sheet: 2 of 21		

[3] Project Architecture



Comments:	Company: EPFL Xplore Research			Variant:	Git Hash: 04d3b06
	Board Name: Amulet Motion Controller			Project Name: Chienpanzé	
Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-22	Revision: 1.1	
Sheet Path: /Project Architecture/		Reviewer:	Size: A3	Sheet: 3 of 21	

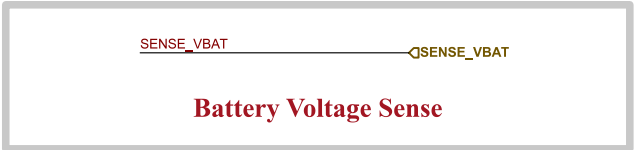
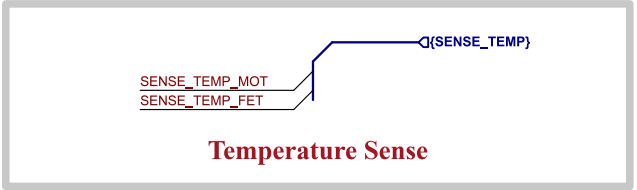
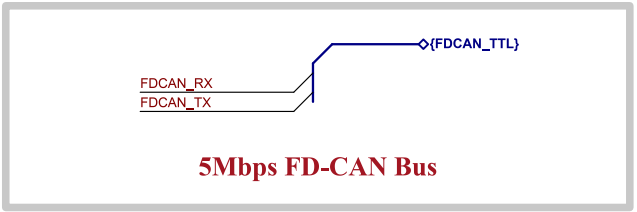
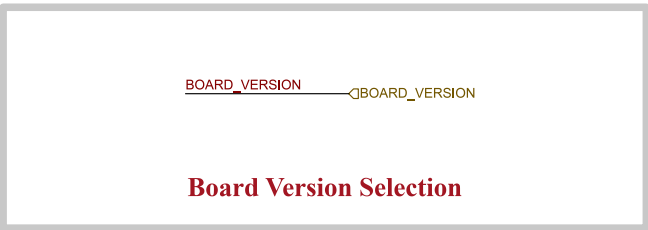
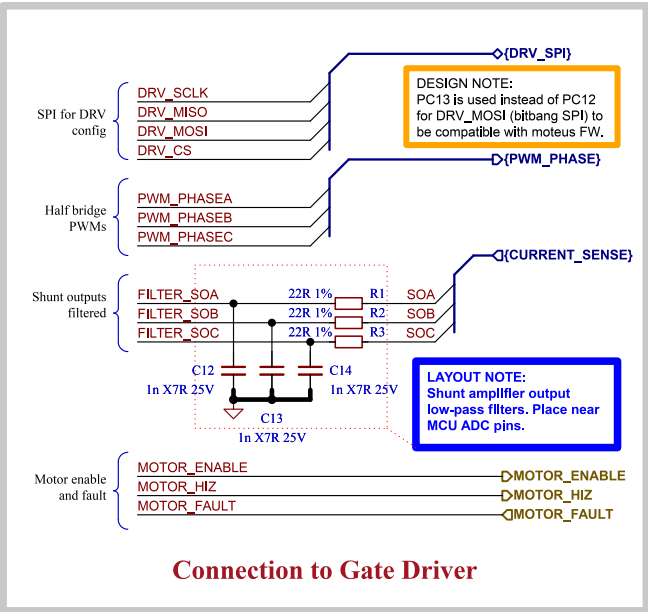
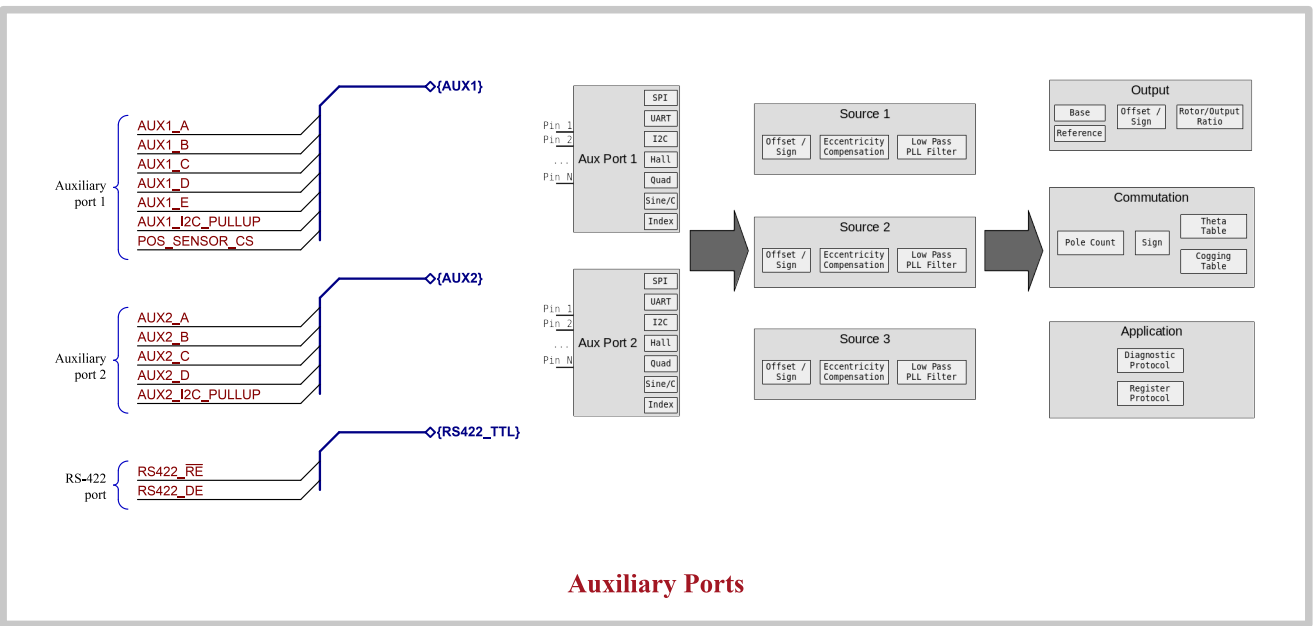
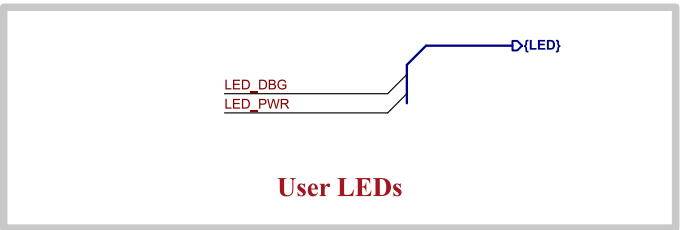
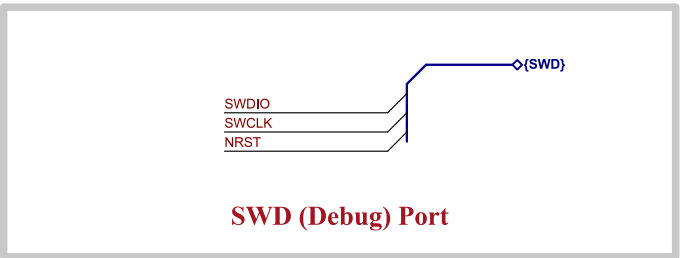
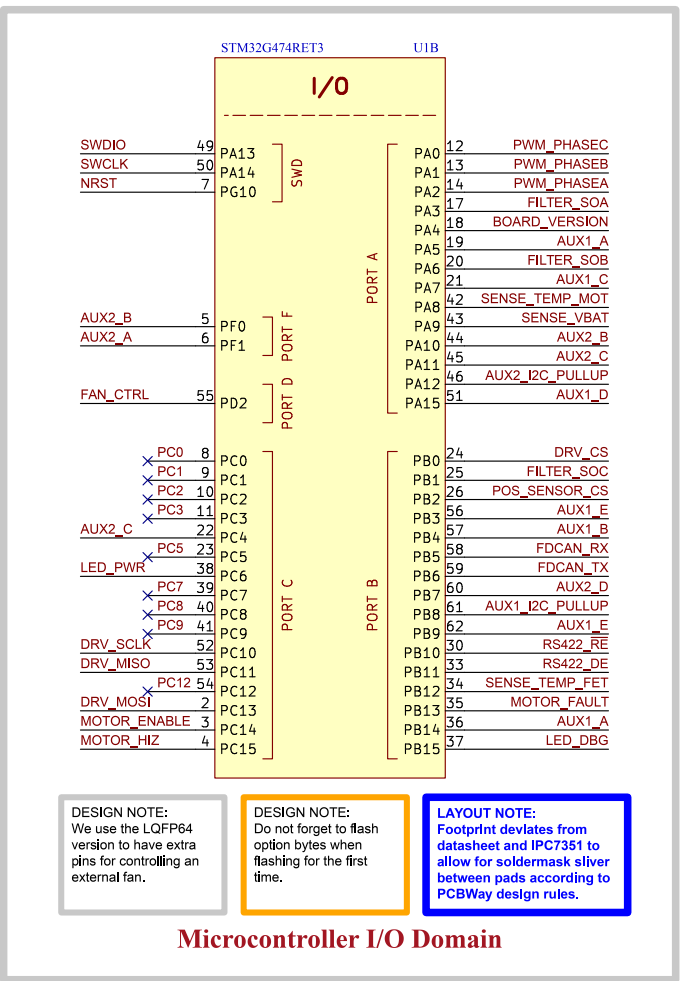
[4] MCU - Power



Microcontroller Power Supply Scheme

	Comments: AN5346 STM32G474 Datasheet p.81 J. Pieper ADC investigation		Company: EPFL Xplore Research		Variant:	Git Hash: 04d3b06
			Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: MCU - Power		File Name: MCU - Power.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-18	Revision: 1.1
	Sheet Path: /Project Architecture/MCU - Power/			Reviewer:	Size: A4	Sheet: 4 of 21

[5] MCU - I/Os



Gate Driver

MOTOR_ENABLE	TP1	MOTOR_ENABLE
MOTOR_HI_Z	TP2	MOTOR_HI_Z
MOTOR_FAULT	TP3	MOTOR_FAULT
DRV_SCLK	TP4	DRV_SCLK
DRV_MISO	TP5	DRV_MISO
DRV_MOSI	TP6	DRV_MOSI
DRV_CS	TP7	DRV_CS
PWM_PHASEA	TP8	PWM_PHASEA
PWM_PHASEB	TP9	PWM_PHASEB
PWM_PHASEC	TP10	PWM_PHASEC
SOA	TP11	SOA
SOB	TP12	SOB
SOC	TP13	SOC

Debug

SWDIO	TP14	SWDIO
SWCLK	TP15	SWCLK
NRST	TP16	NRST

Auxiliary pins 1

AUX1_A	TP17	AUX1_A
AUX1_B	TP18	AUX1_B
AUX1_C	TP19	AUX1_C
AUX1_D	TP20	AUX1_D
AUX1_E	TP21	AUX1_E
AUX1_I2C_PULLUP	TP22	AUX1_I2C_PULLUP
POS_SENSOR_CS	TP23	POS_SENSOR_CS

Auxiliary pins 2

AUX2_A	TP24	AUX2_A
AUX2_B	TP25	AUX2_B
AUX2_C	TP26	AUX2_C
AUX2_D	TP27	AUX2_D
AUX2_I2C_PULLUP	TP28	AUX2_I2C_PULLUP

RS-422

RS422_RE	TP29	RS422_RE
RS422_DE	TP30	RS422_DE

LEDs

LED_DBG	TP31	LED_DBG
LED_PWR	TP32	LED_PWR

FD-CAN

FDCAN_RX	TP33	FD_CAN_RX
FDCAN_TX	TP34	FDCAN_TX

Fan

FAN_CTRL	TP35	FAN_CTRL
----------	------	----------

Sense

SENSE_TEMP_MOT	TP36	SENSE_TEMP_MOT
SENSE_TEMP_FET	TP37	SENSE_TEMP_FET
SENSE_VBAT	TP38	SENSE_VBAT

Test Points

Comments:

References:

Flexible I/O worked examples
Flexible I/O source configuration

Sheet Title:

MCU - I/Os

Sheet Path:

/Project Architecture/MCU - I/Os/

Company:

EPFL Xplore Research

Board Name:

Amulet Motion Controller

File Name:

MCU - I/Os.kicad_sch

Designer:

Vincent Nguyen

Reviewer:

Variant:

EPFL Xplore Research

Project Name:

Chienpanzé

Date:

2023-12-20

Size:

A3

Git Hash:

04d3b06

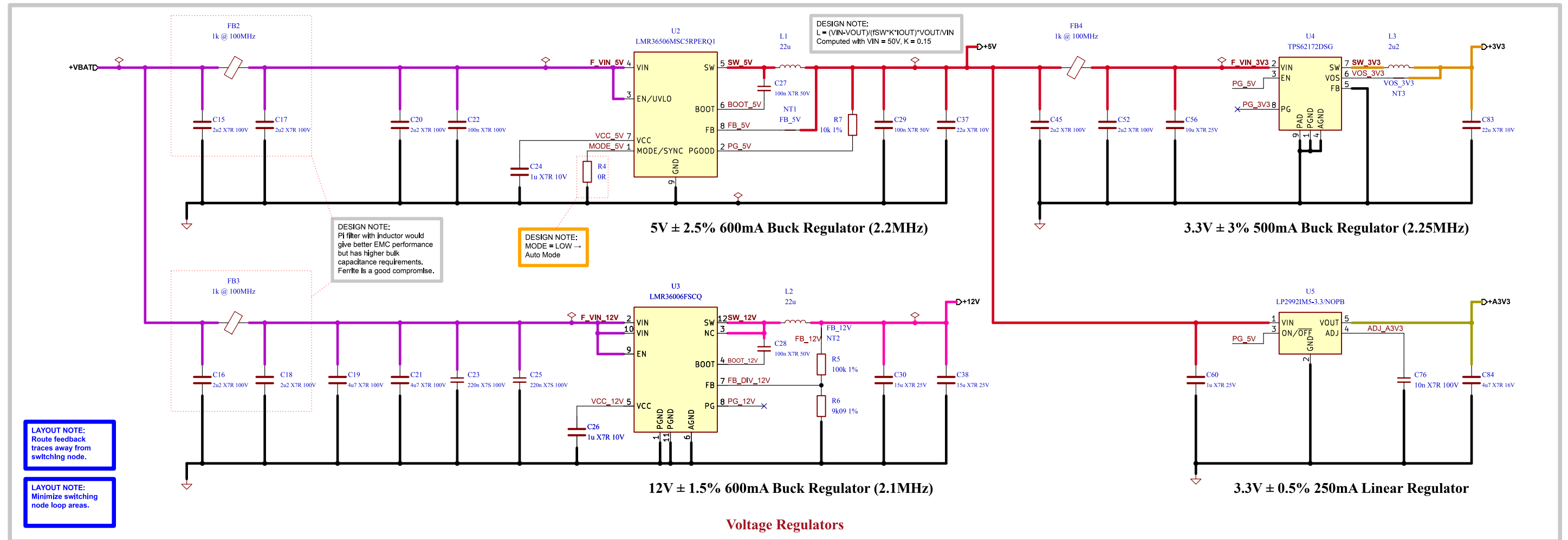
Revision:

1.1

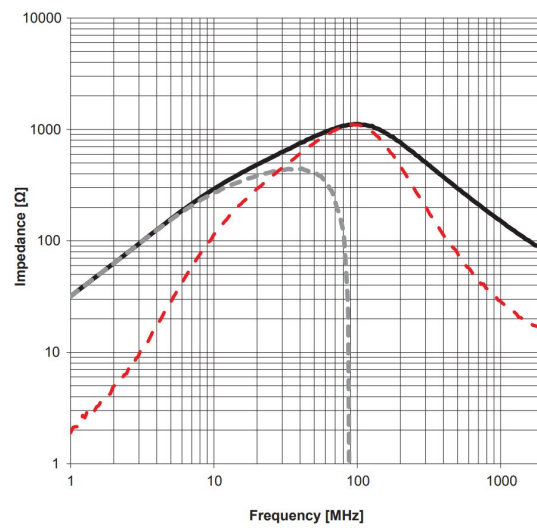
Sheet:

5 of 21

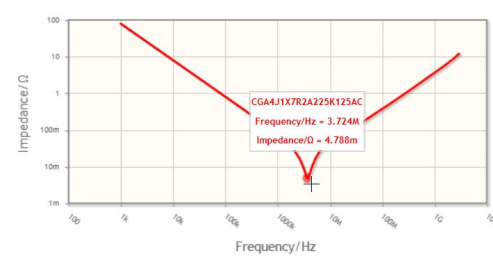
[6] Power - Generation



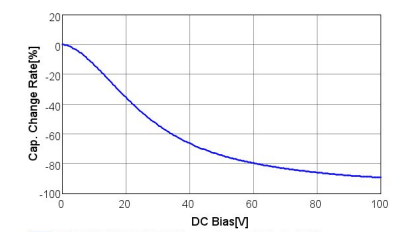
Ferrite bead Impedance vs Frequency curve



Pi filter capacitor Impedance vs Frequency curve



Bulk capacitance Derating curve



Capacitor and Ferrite Bead Parameters

Battery

+VBAT TP39 +VBAT
+VBAT TP40 +VBAT
+VBAT TP41 +VBAT

Filtered Power Rails

F_VIN_12V TP42 F_VIN_12V
F_VIN_5V TP43 F_VIN_5V
F_VIN_3V3 TP44 F_VIN_3V3

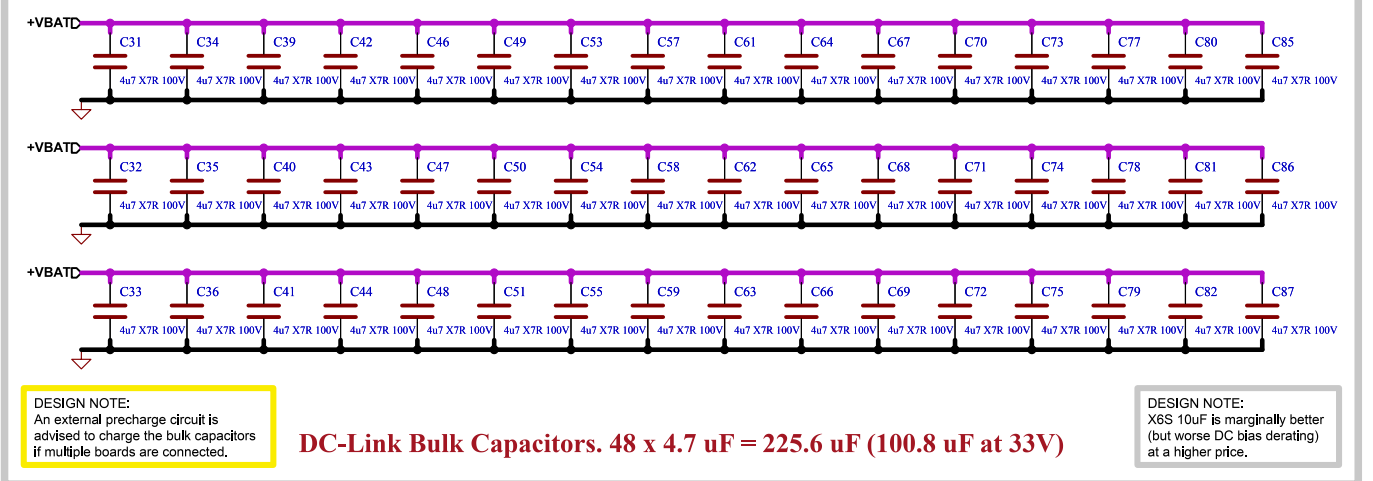
Power Rails

+12V TP45 +12V
+5V TP46 +5V
+5V TP47 +5V
+3V3 TP48 +3V3
+3V3 TP49 +3V3
+3V3 TP50 +3V3
+A3V3 TP51 +A3V3

Ground

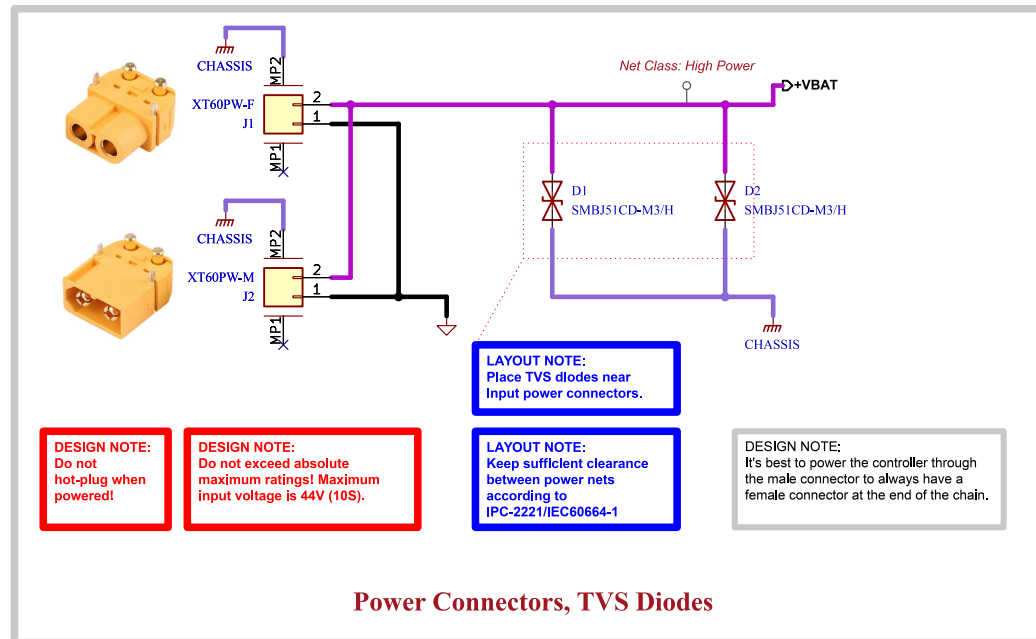
GND TP52 GND
GND TP53 GND
GND TP54 GND
GND TP55 GND
GND TP56 GND
GND TP57 GND
GND TP58 GND

Test Points



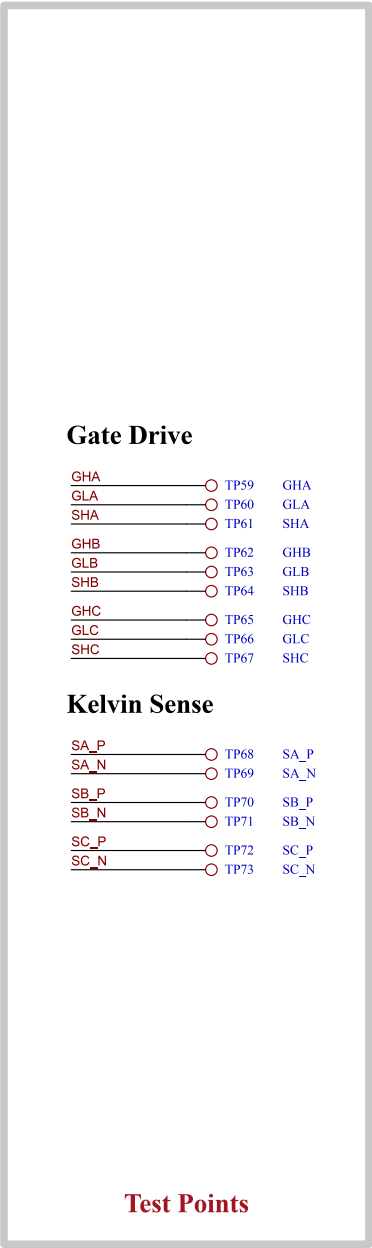
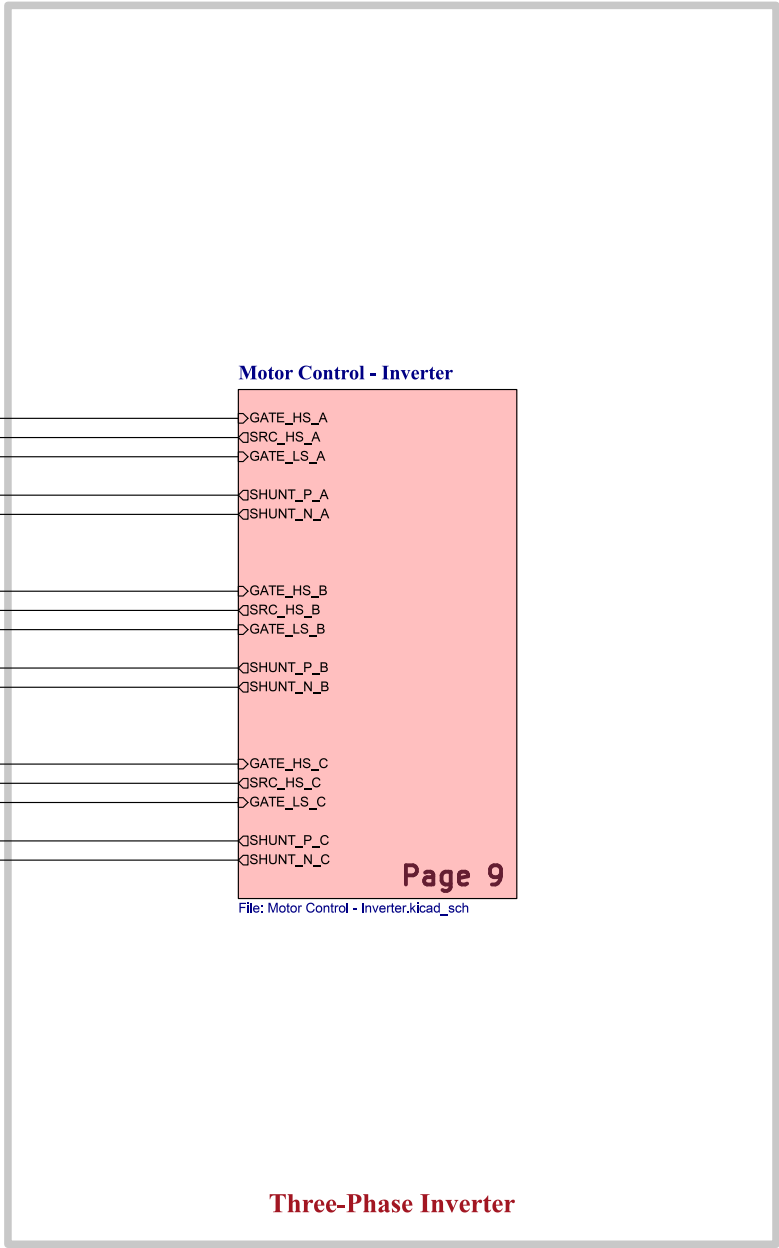
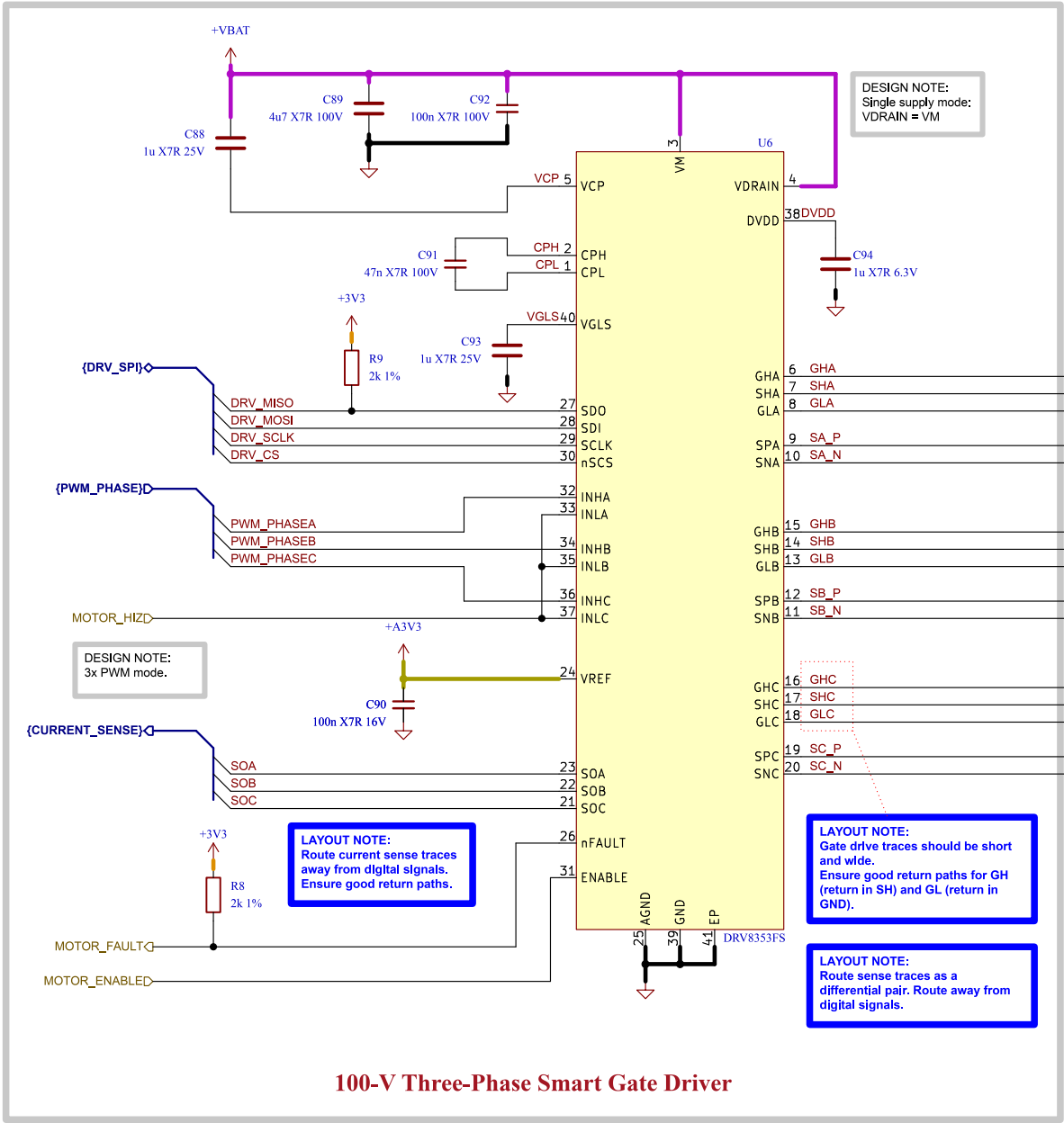
Comments: +12V (600 mA): Fan power +5V (600 mA): Gate Driver, LEDs +3V3 (500 mA): digital 3.3V +3V3A (250 mA): analog 3.3V	Company: EPFL Xplore Research		Variant:	Git Hash: 04d3b06
	Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
Sheet Title: Power - Generation	File Name: Power - Generation.kicad_sch	Designer: Vincent Nguyen	Date: 2024-04-13	Revision: 1.1
Sheet Path: /Project Architecture/Power - Generation/	Reviewer:		Size: A3	Sheet: 6 of 21

[7] Power - Connectors



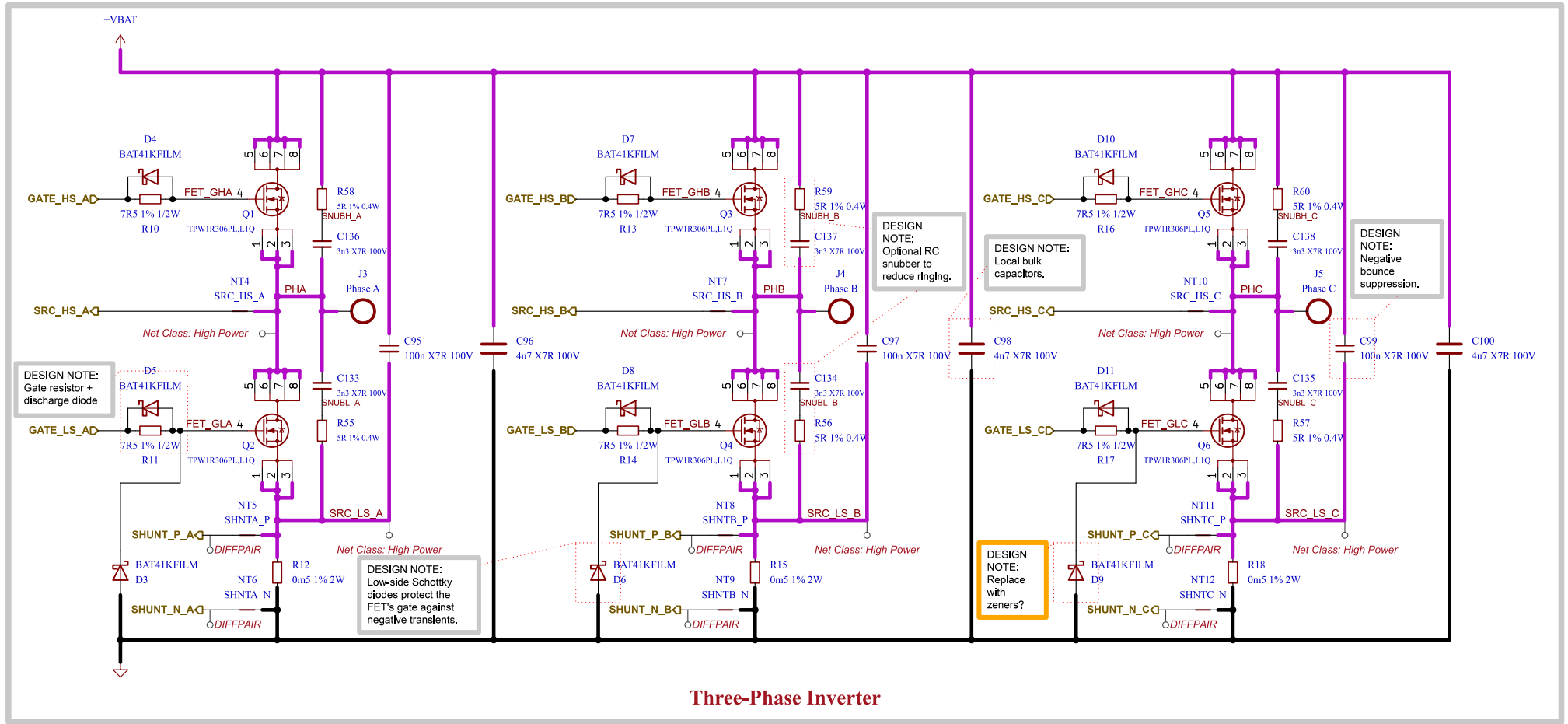
	Comments:	Company: EPFL Xplore Research		Variant:	Git Hash: 04d3b06
		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Power - Connectors	File Name: Power - Connectors.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-31	Revision: 1.1
	Sheet Path: /Project Architecture/Power - Connectors/		Reviewer:	Size: A4	Sheet: 7 of 21

[8] Motor Control - Top Level



	Comments:		Company:			Variant:		Git Hash:	
			EPFL Xplore Research					04d3b06	
			Board Name:			Project Name:			
			Amulet Motion Controller			Chienpanzé			
	Sheet Title:		File Name:		Designer:		Date:		Revision:
Motor Control - Top Level		Motor Control - Top Level.kicad_sch		Vincent Nguyen		2023-12-20		1.1	
Sheet Path:				Reviewer:		Size:		Sheet:	
/Project Architecture/Motor Control - Top Level/						A3		8 of 21	

[9] Motor Control - Inverter



LAYOUT NOTE:
High current traces must be carefully designed. Ensure ground return path does not cross sensitive parts of the board. Use multiple planes for higher current carrying capacity.

LAYOUT NOTE:
Keep sufficient clearance between power nets according to IPC-2221/IEC60664-1.

DESIGN NOTE:
A gate drive current that is too large can damage the FETs!

Comments:

System Design Considerations for High-Power Motor Driver Applications
Best Practices for Board Layout of Motor Drivers
Proper RC Snubber Design for Motor Drivers

Sheet Title:

Motor Control - Inverter

Sheet Path:

/Project Architecture/Motor Control - Top Level/Motor Control - Inverter/

Company:

EPFL Xplore Research



Variant:

Git Hash:

04d3b06

Board Name:

Amulet Motion Controller

Project Name:

Chienpanzé

File Name:

Motor Control - Inverter.kicad_sch

Designer:

Vincent Nguyen

Date:

2024-01-25

Revision:

1.1

Reviewer:

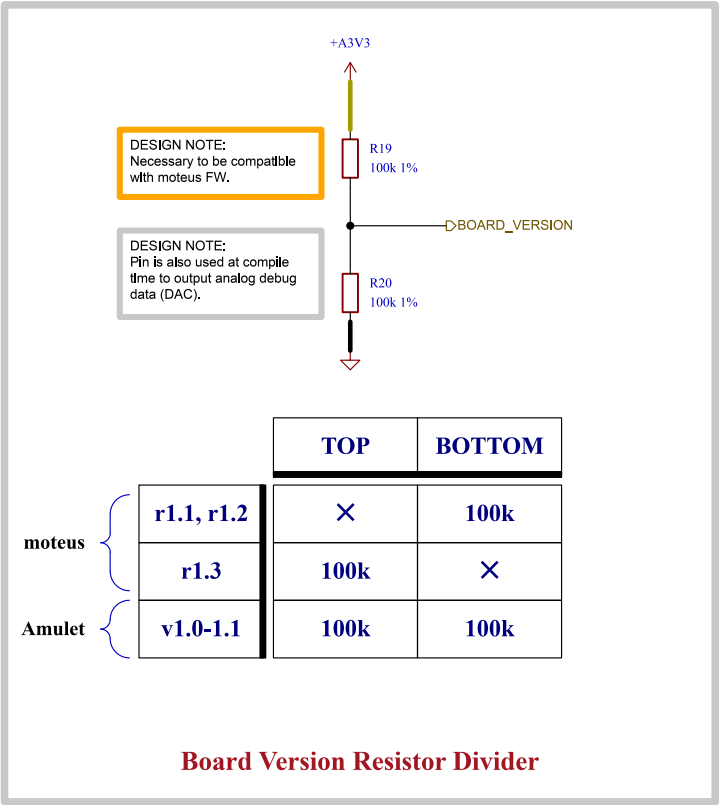
Size:

A4

Sheet:

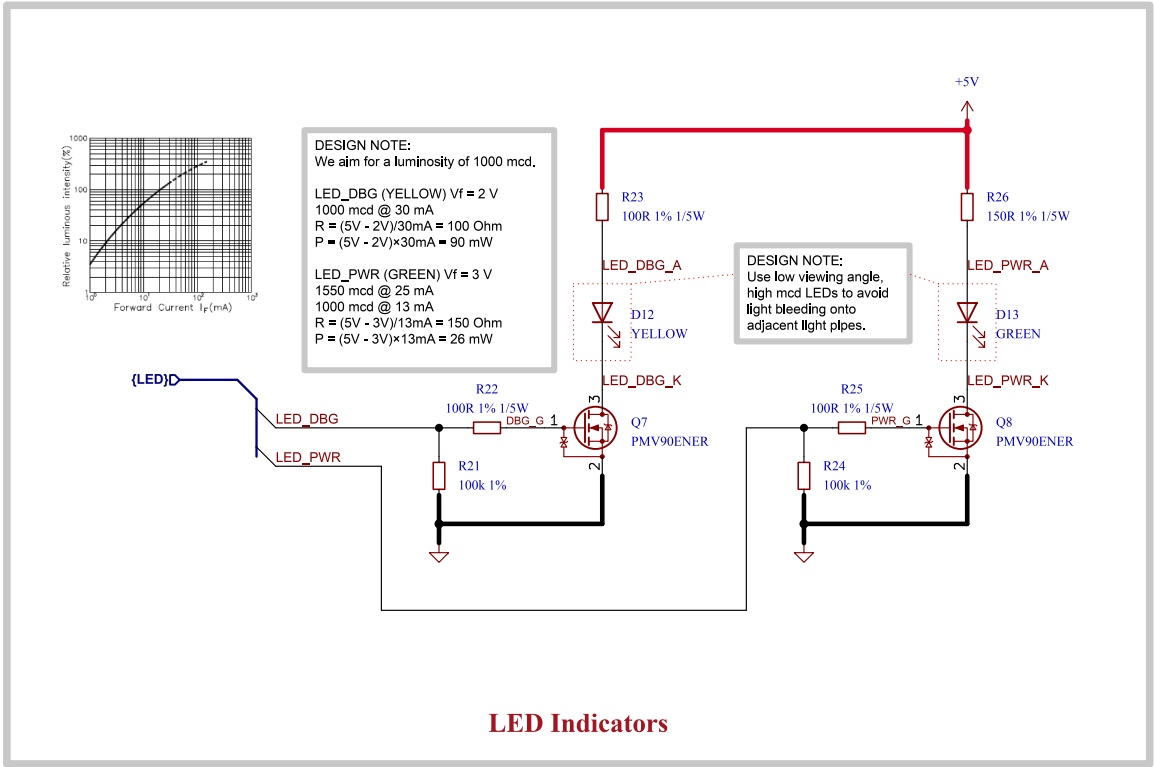
9 of 21

[10] Misc - Board Version, DAC



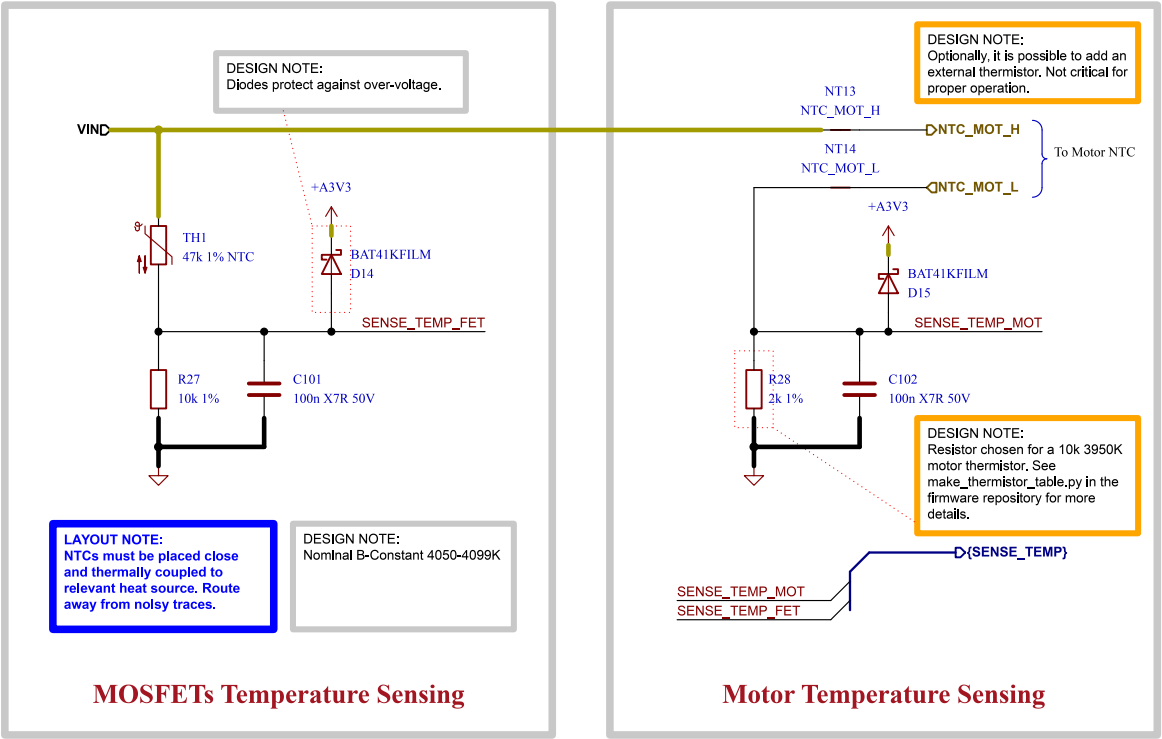
	Comments:	Company: EPFL Xplore Research 		Variant:	Git Hash: 04d3b06
		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Misc - Board Version, DAC	File Name: Misc - Board Version DAC.kicad_sch	Designer: Vincent Nguyen	Date: 2024-04-13	Revision: 1.1
	Sheet Path: /Project Architecture/Misc - Board Version, DAC/		Reviewer:	Size: A4	Sheet: 10 of 21

[11] User - LED Indicators



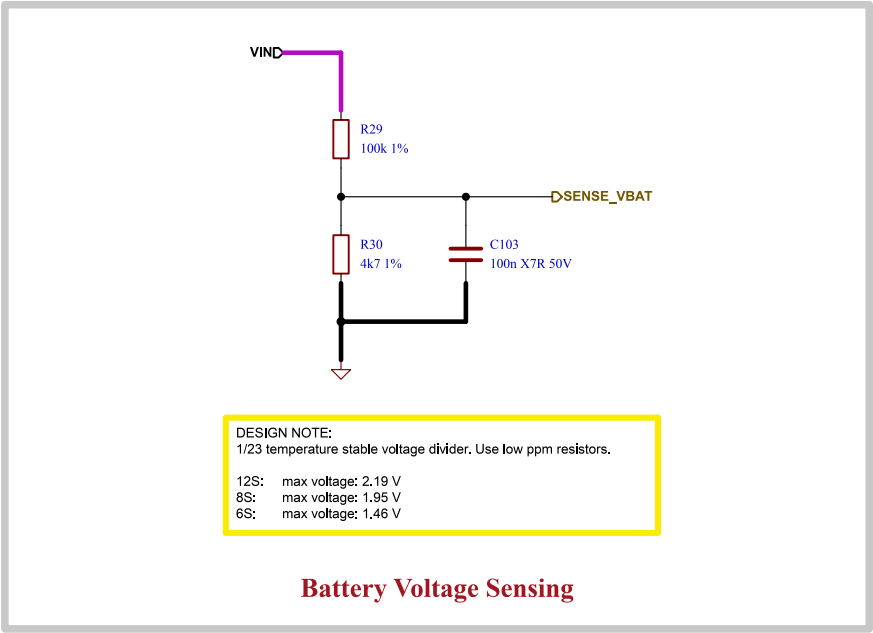
	Comments:	Company: EPFL Xplore Research		Variant:	Git Hash: 04d3b06
		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: User - LED Indicators	File Name: User - LED Indicators.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-19	Revision: 1.1
	Sheet Path: /Project Architecture/User - LED Indicators/		Reviewer:	Size: A4	Sheet: 11 of 21

[12] Sensing - Temperature



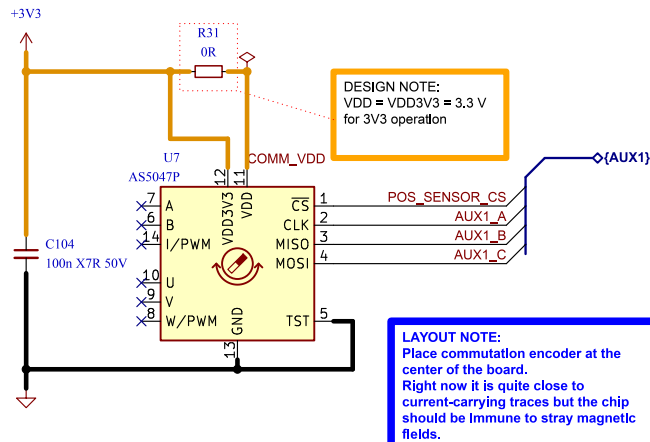
	Comments:	Company:			Variant:	Git Hash:	
		EPFL Xplore Research			04d3b06		
		Board Name:				Project Name:	
		Amulet Motion Controller				Chienpanzé	
Sheet Title:	File Name:	Designer:	Date:	Revision:			
Sensing - Temperature	Sensing - Temperature.kicad_sch	Vincent Nguyen	2024-04-13	1.1			
Sheet Path:		Reviewer:	Size:	Sheet:			
/Project Architecture/Sensing - Temperature/			A4	12 of 21			

[13] Sensing - Battery



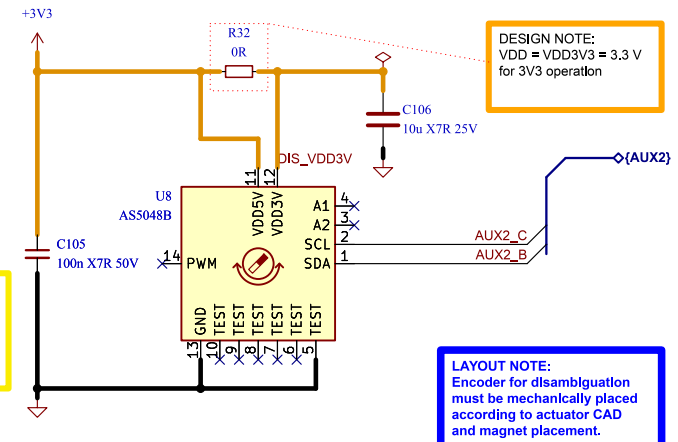
	Comments:	Company: EPFL Xplore Research		Variant:	Git Hash: 04d3b06
		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Sensing - Battery	File Name: Sensing - Battery.kicad_sch	Designer: Vincent Nguyen	Date: 2023-10-14	Revision: 1.1
	Sheet Path: /Project Architecture/Sensing - Battery/		Reviewer:	Size: A4	Sheet: 13 of 21

[14] Sensing - Position



14-Bit On-Axis Magnetic Rotary Position Sensor (commutation)

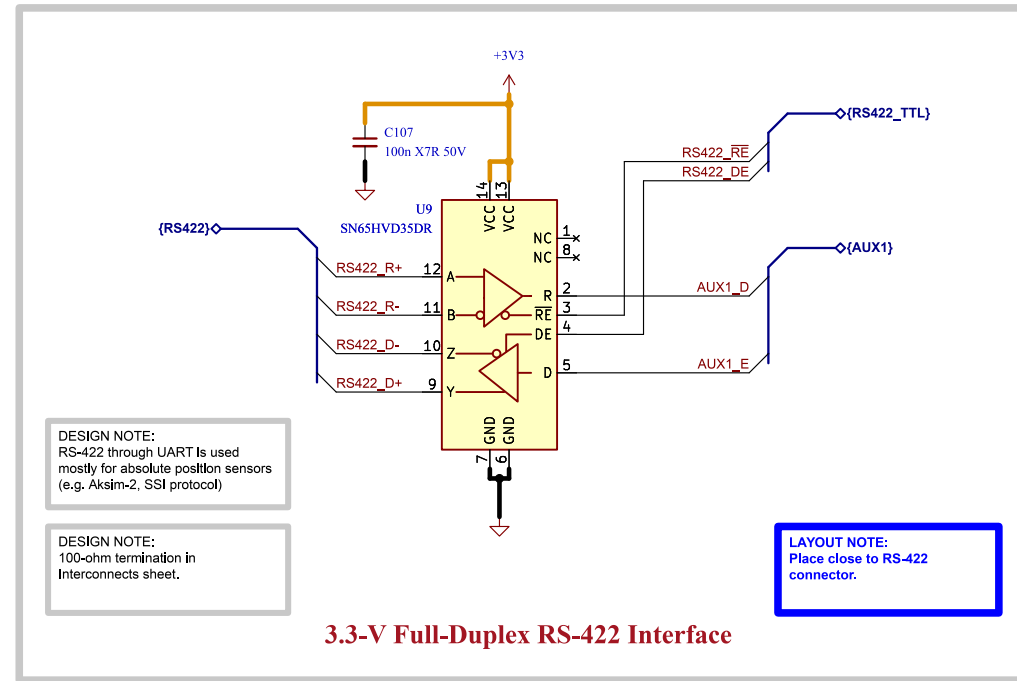
DESIGN NOTE:
AS5047P senses magnet mounted on planetary sun gear, for commutation.
AS5048B senses magnet mounted on shaft with same reduction factor as planetary gearbox for disambiguation.



14-Bit On-Axis Magnetic Rotary Position Sensor (disambiguation)

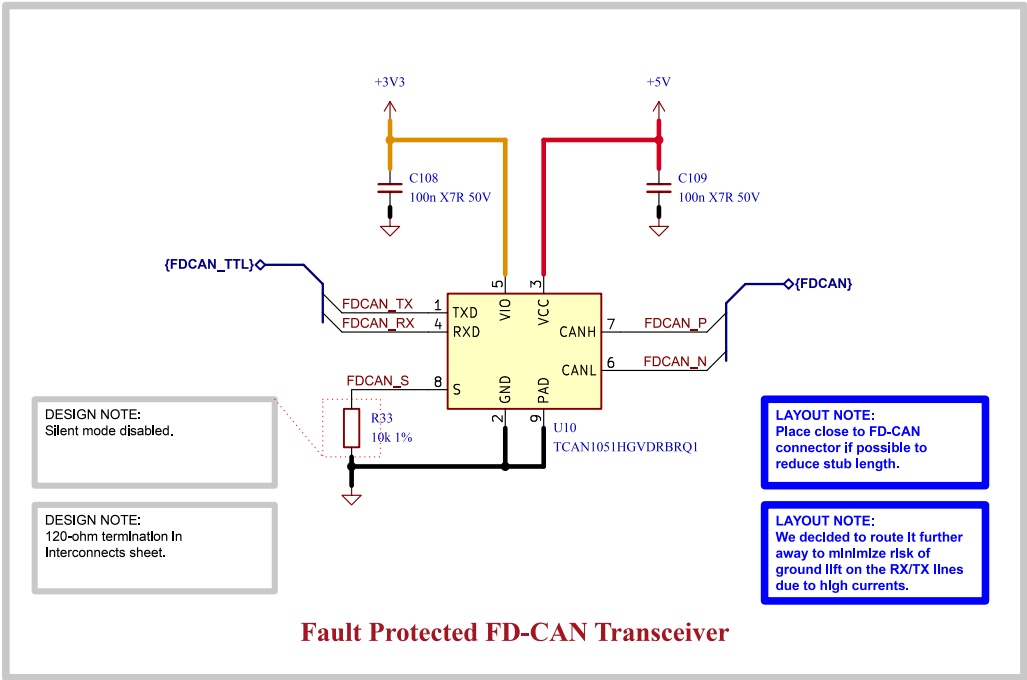
	Comments:	Company: EPFL Xplore Research	Variant:	Git Hash: 04d3b06
		Board Name: Amulet Motion Controller	Project Name: Chienpanzé	
	Sheet Title: Sensing - Position	File Name: Sensing - Position.kicad_sch	Designer: Vincent Nguyen	Date: 2023-10-14
	Sheet Path: /Project Architecture/Sensing - Position/	Reviewer:	Size: A4	Revision: 1.1
			Sheet: 14 of 21	

[15] Interface - RS-422



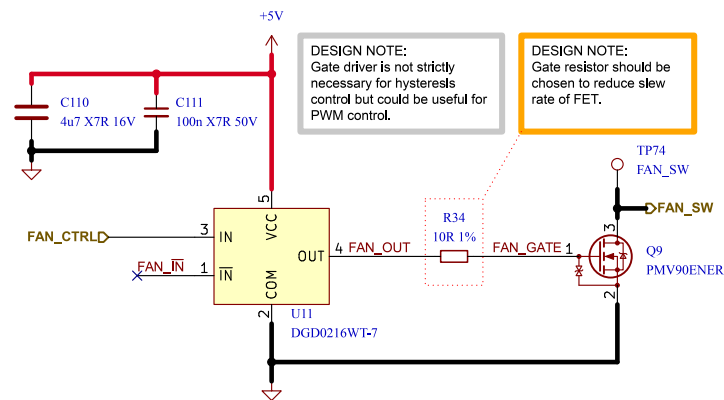
	Comments:	Company: EPFL Xplore Research		Variant:	Git Hash: 04d3b06
		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Interface - RS-422	File Name: Interface - RS-422.kicad_sch	Designer: Vincent Nguyen	Date: 2023-10-15	Revision: 1.1
	Sheet Path: /Project Architecture/Interface - RS-422/		Reviewer:	Size: A4	Sheet: 15 of 21

[16] Interface - FD-CAN



	Comments:	Company: EPFL Xplore Research		Variant:	Git Hash: 04d3b06
		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Interface - FD-CAN	File Name: Interface - FD-CAN.kicad_sch	Designer: Vincent Nguyen	Date: 2023-10-15	Revision: 1.1
	Sheet Path: /Project Architecture/Interface - FD-CAN/		Reviewer:	Size: A4	Sheet: 16 of 21

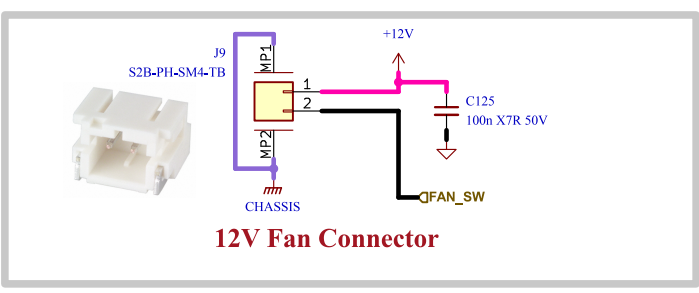
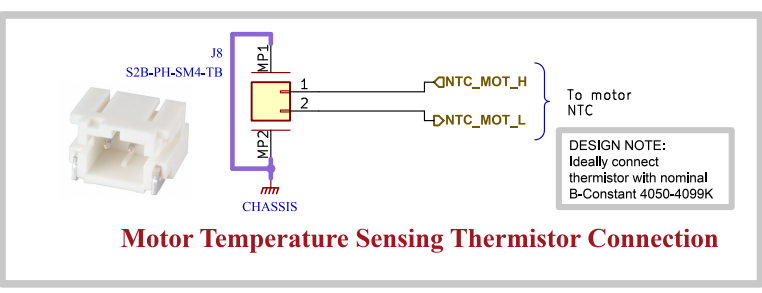
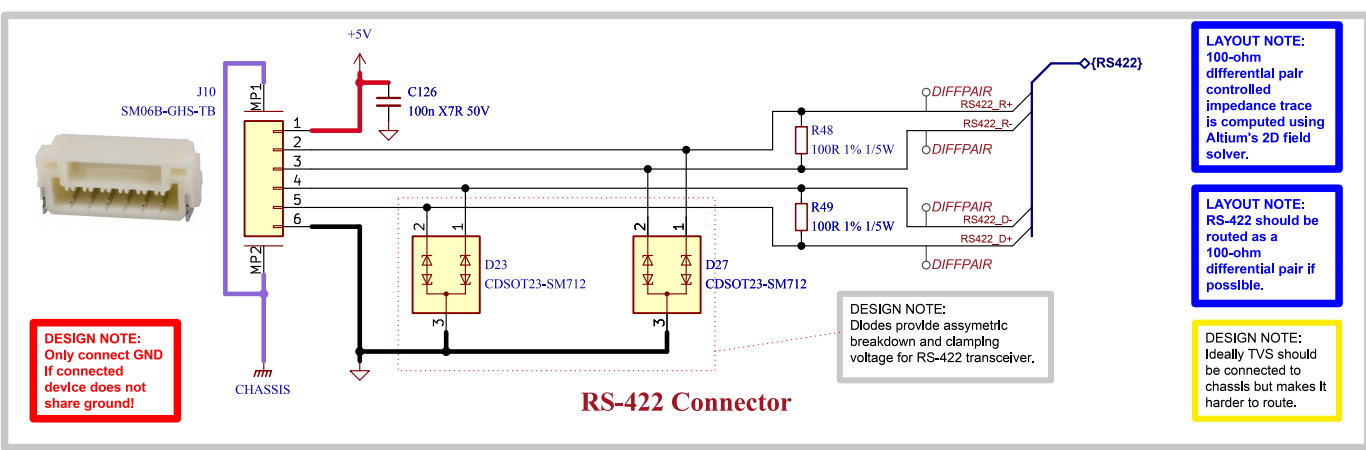
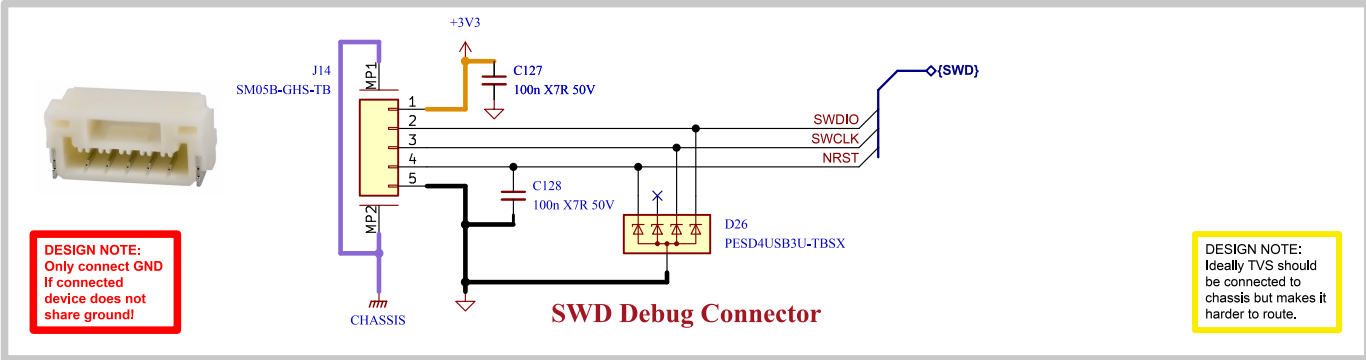
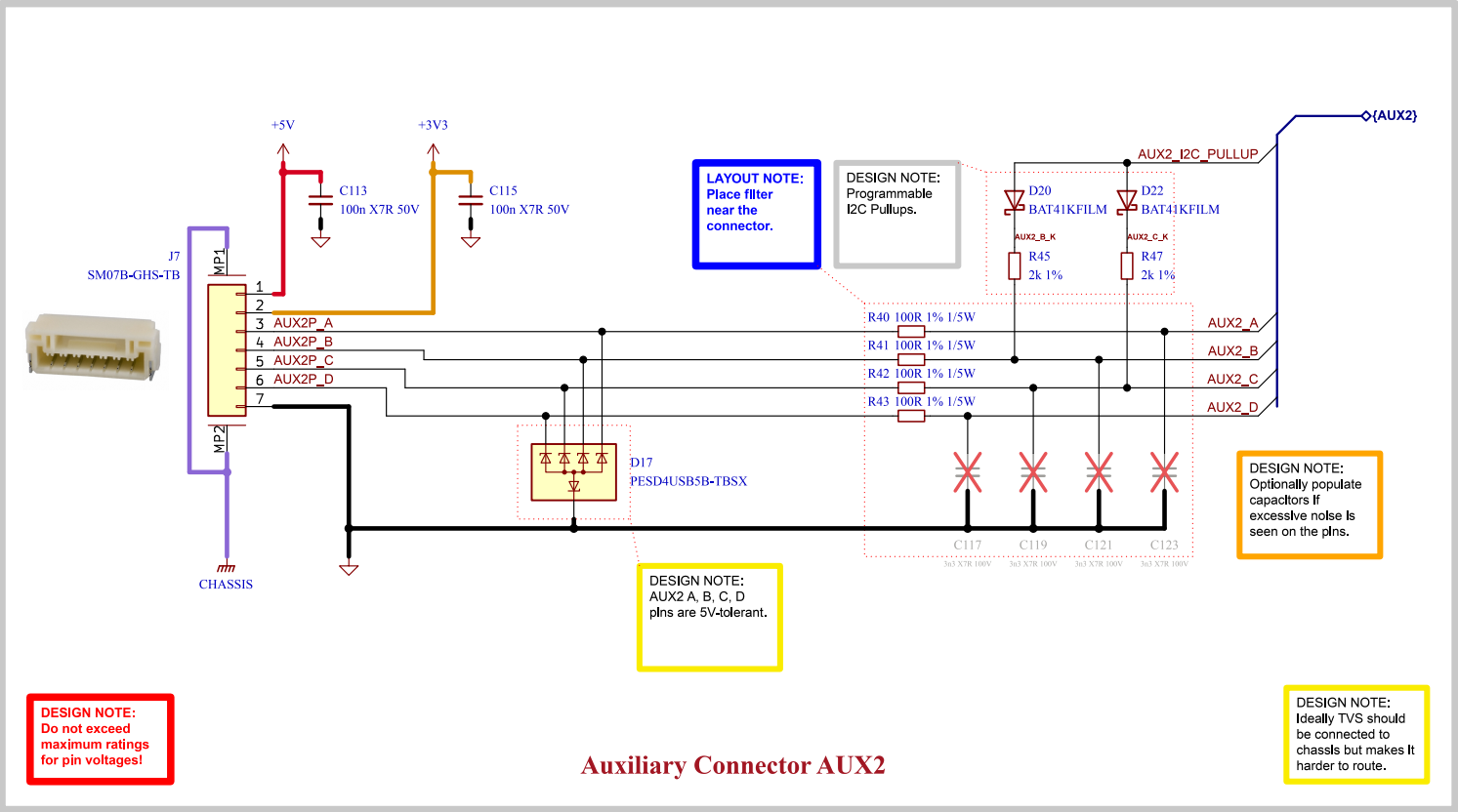
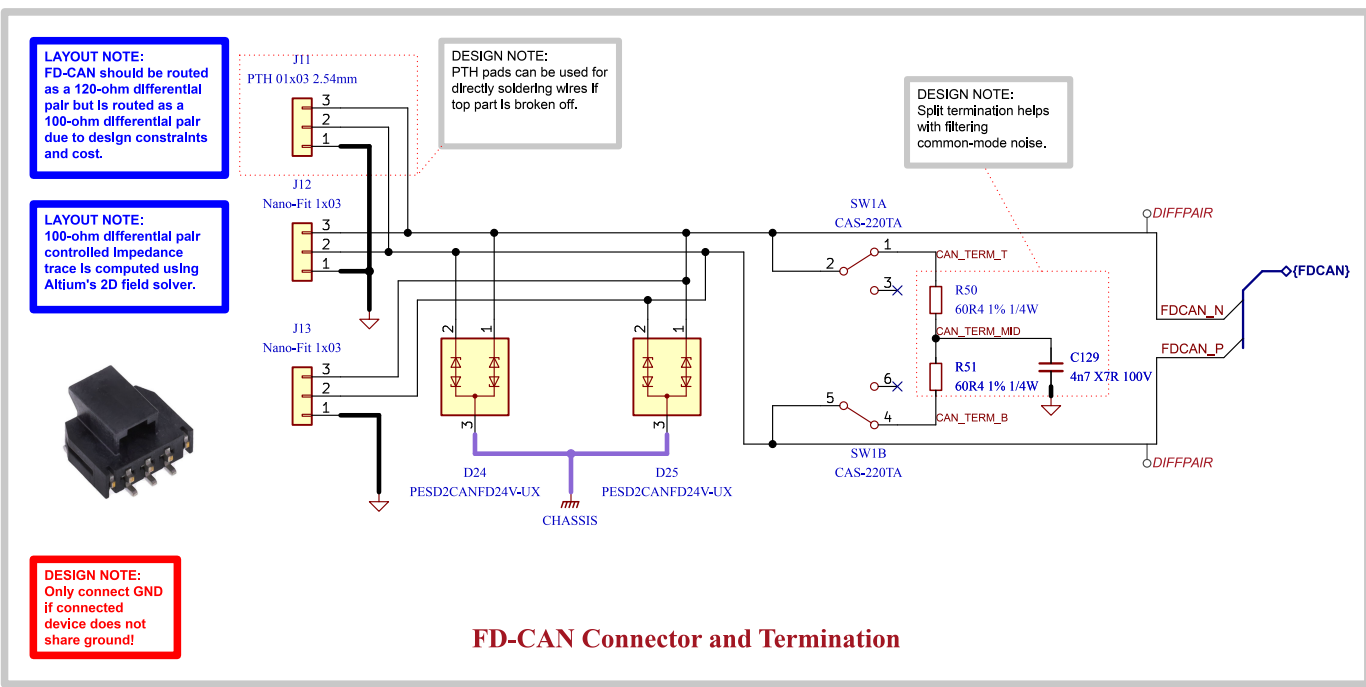
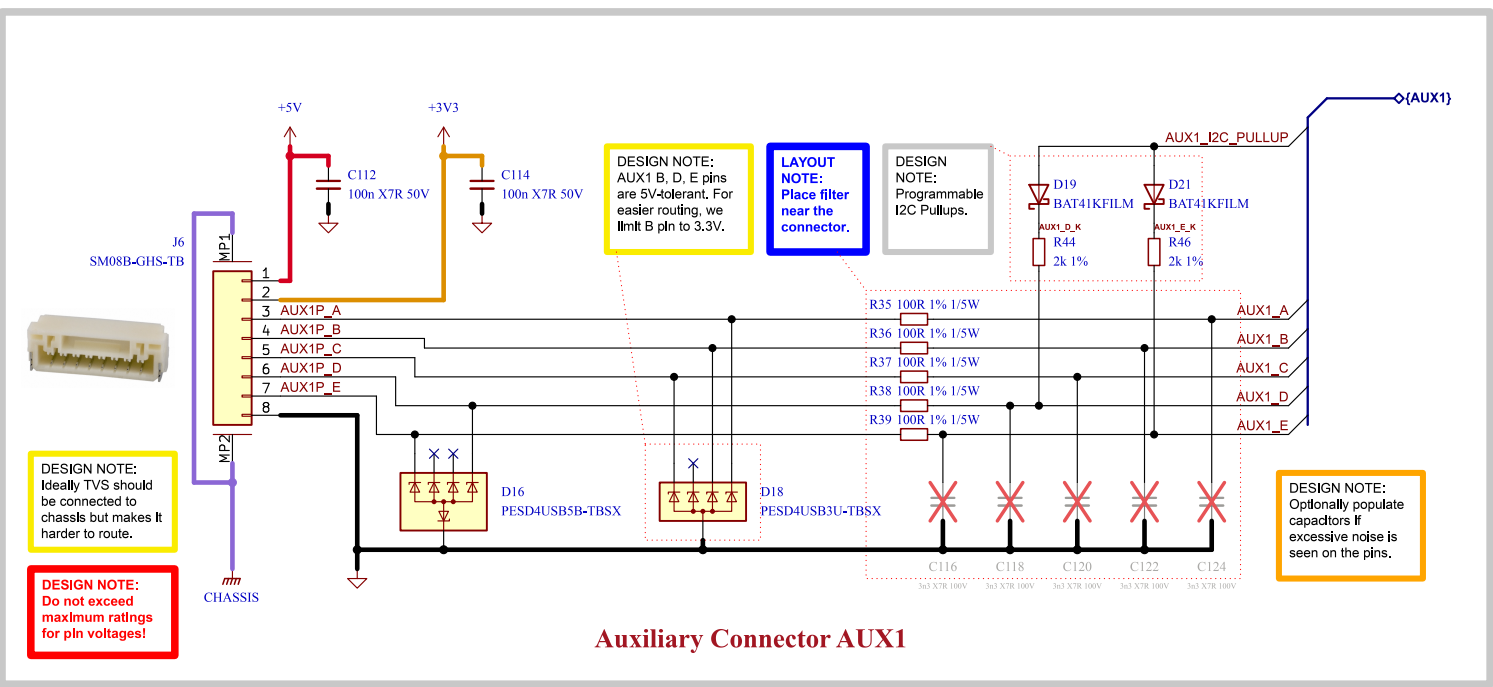
[17] Interface - Fan Control



Low-side Fan Switch and Gate Driver

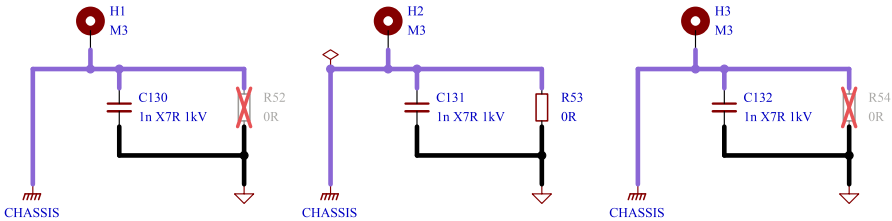
	Comments:	Company: EPFL Xplore Research		Variant:		Git Hash: 04d3b06	
		Board Name: Amulet Motion Controller			Project Name: Chienpanzé		
	Sheet Title: Interface - Fan Control	File Name: Interface - Fan Control.kicad_sch	Designer: Vincent Nguyen		Date: 2023-11-19	Revision: 1.1	
Sheet Path: /Project Architecture/Interface - Fan Control/		Reviewer:		Size: A4	Sheet: 17 of 21		

[18] Interface - Interconnects



	Comments: Reference: Flexible I/O worked examples	Company: EPFL Xplore Research 		Variant:		Git Hash: 04d3b06	
		Board Name: Amulet Motion Controller			Project Name: Chienpanzé		
	Sheet Title: Interface - Interconnects	File Name: Interface - Interconnects.kicad_sch	Designer: Vincent Nguyen		Date: 2023-12-31	Revision: 1.1	
	Sheet Path: /Project Architecture/Interface - Interconnects/			Reviewer:		Size: A3	Sheet: 18 of 21

[19] Misc - Holes, Fiducials



DESIGN NOTE:
Optional RC filter for chassis
grounding. Capacitors offer multi-point
low-impedance connections at
high-frequencies.

DESIGN NOTE:
0-Ohm resistor provides low-impedance
bond between chassis and signal GND.
Be aware of ground loops through cable
shield.

Mounting Holes and Chassis Connection

Front



Back



Fiducial Markers

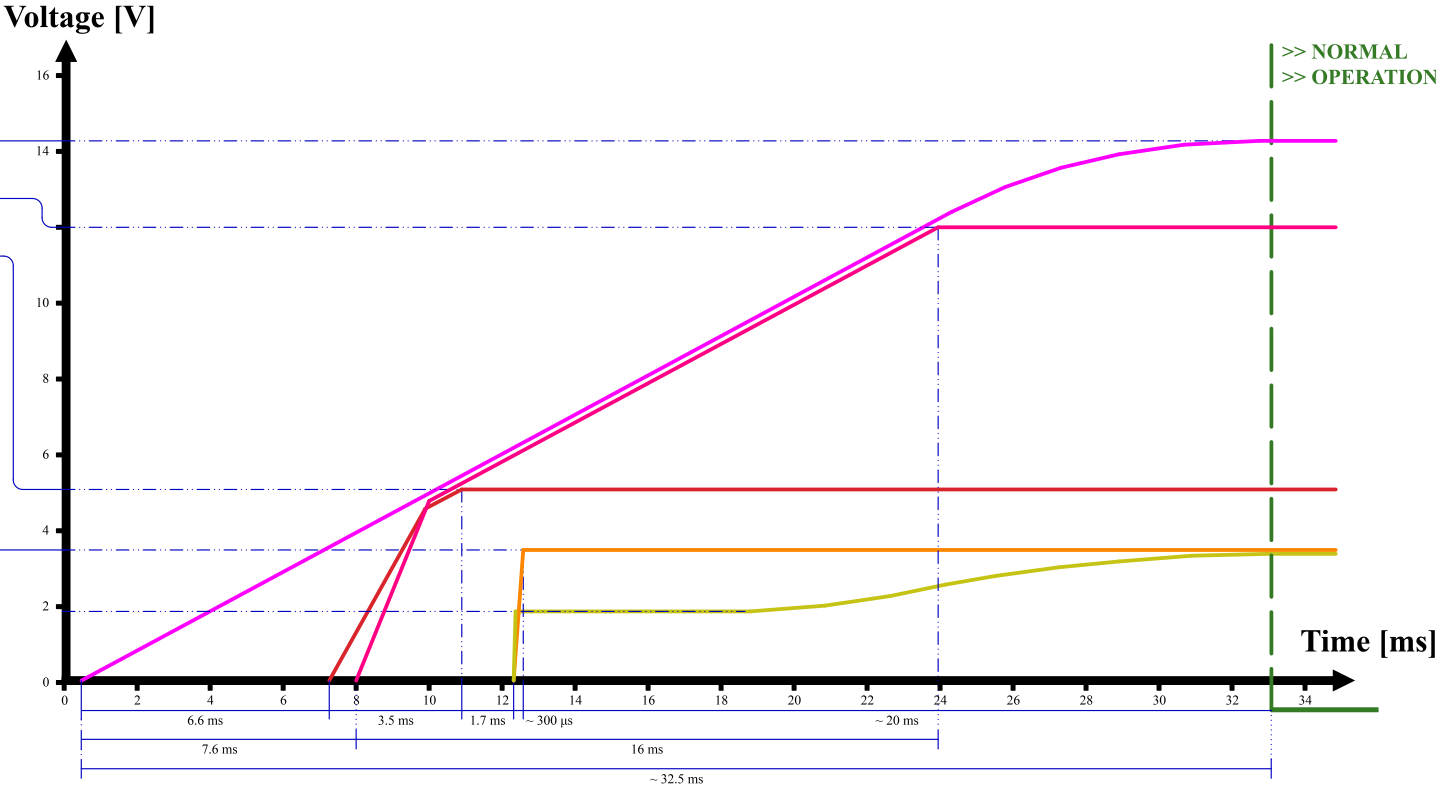
	Comments:	Company: EPFL Xplore Research				Variant:	Git Hash: 04d3b06	
		Board Name: Amulet Motion Controller				Project Name: Chienpanzé		
	Sheet Title: Misc - Holes, Fiducials	File Name: Misc - Holes Fiducials.kicad_sch	Designer: Vincent Nguyen		Date: 2023-10-22		Revision: 1.1	
	Sheet Path: /Project Architecture/Misc - Holes, Fiducials/		Reviewer:		Size: A4	Sheet: 19 of 21		

[20] Power - Sequencing

NAME	SOURCE	LEVEL
+VBAT	BATTERY	12 - 44V
+12V	LMR36006	12V $\pm$ 1.5%
+5V	LMR36506	5V $\pm$ 1.5%
+3V3	TPS62172	3.3V $\pm$ 3%
+A3V3	LP2992	3.3V $\pm$ 0.5%

DESIGN NOTE:
Power sequencing graph measured experimentally.

DESIGN NOTE:
Start-up time of 12V rail depends on +VBAT charging time, which itself depends on the precharge circuit.



	Comments:	Company: EPFL Xplore Research		Variant:	Git Hash: 04d3b06
		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Power - Sequencing	File Name: Power - Sequencing.kicad_sch	Designer: Vincent Nguyen	Date: 2024-03-12	Revision: 1.1
	Sheet Path: /Power - Sequencing/		Reviewer:	Size: A4	Sheet: 20 of 21

[21] Revision History

12-DEC-2023 - Initial Release
Variant: v1.0 Preliminary

- Changed CPH-CPL capacitor to 47nF (gate driver).
- Changed FD-CAN transceiver IC.
- Changed FETs for top cooled variant.
- Added TVS protection and termination switch to FD-CAN.
- Added low-side switched 12V 600mA source for external fan.
- Added LDO for analog supply.
- Changed input power TVS diode to bidirectional and added one diode per connector.
- Moved SOx low-pass filter to MCU section. Should be placed near MCU to avoid noise coupling into ADC lines.
- Added second onboard I2C magnetic encoder for disambiguation.
- Switched PWM_PHASEA with PWM_PHASEC on STM32G474 pinout for easier routing.
- Changed RS422 pinout on connector.
- Added ESD protection to all interfaces.
- Added overvoltage protection on thermistor ADC inputs.
- Changed buck regulators to optimize for low noise.
- Added Pi filters to inputs of buck regulators and MCU analog supply.
- Added decoupling caps next to power pins of connectors.

25-JAN-2024 - First Revision
Variant: v1.0 Checked

- Added controller target specifications.
- Replaced 5V 300mA buck converter with 600mA version.
- Added credits to moteus on cover page.
- Added optional RC-Snubber to power stage.
- Increased chassis length to go around the board.
- CAN and power TVS diodes now go to chassis.
- Changed clearance between nets to respect IEC60664-1 where possible.
- Rectified comment on precharge.
- Changed power TVS diode reference designator from "U" to "D".
- Replaced chassis-GND capacitor by 1nF 1kV.

12-MAR-2024 - First Revision
Variant: v1.0 Released

- [illegible]

13-APR-2024 - Second Revision
Variant: v1.1 Released

- [illegible]

	Comments:	Company: EPFL Xplore Research 		Variant:	Git Hash:
		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Vincent Nguyen	Date: 2024-01-03	Revision: 1.1
	Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 21 of 21